High Powered Rocket Datalogger

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Context

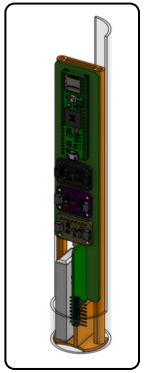
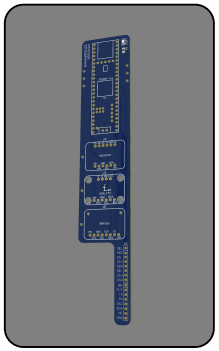
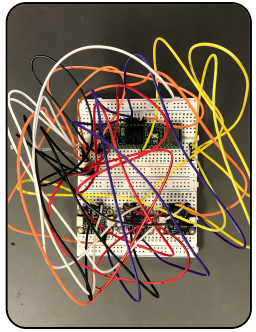
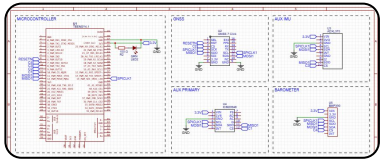
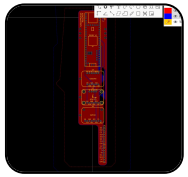
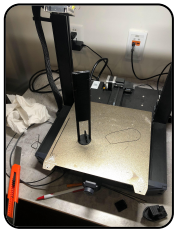
- Space Grant Midwest High Powered Rocket Competition 2025
- Two competition performance goals are maximum altitude and hitting a 1000 ft target altitude at apogee
- Another requirement for non-commercial sensor suite for data logging
- Team chose to pursue a minimum diameter rocket (2” OD)

Role

- In charge of designing non-commercial sensor suite

Development

- Selection of components done through research and seeing what other teams have done in the past
- Biggest challenge was fitting components in the tube due to the minimum diameter requirement, which limited what boards I could use
- Chose Teensy 4.1 for microcontroller, ADXL375 IMU to handle high G loads, ICM20948 for main IMU, u-blox NEO10M for GPS, and BMP390 for barometer
- First draft of PCB - NDSU did not have the capability to help us with soldering chips, and determined it would be easier to use breakout boards
- Second draft - Used breakout boards for easier mounting of components
- Custom PCBS designed through EasyEDA then sent to JLCPCB for manufacturing

Outcomes

- Team was in 3rd place leading up to competition
- Sensor suite was not ready for test launch and did not see action
- Rocket crashed before any more progress was mode. Project closed out post crash
- Learned how to interface with a Teensy in VSCode through platformio, and how to create electrical diagrams